CALCULUS · THE FUNDAMENTAL THEOREM OF CALCULUS

Accumulation Function Explorer

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Accumulation Function Explorer** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. If $f(t) > 0$ on an interval, is $A(x) = \int_0^x f(t)\,dt$ increasing or decreasing there?
   Increasing: positive area is being added.
2. Where do you think A has a maximum?
   Where f changes from positive to negative.

2 Explore use the tool

1. $f(t) = 2 - t$ with $a = 0$: record A(1), A(2), A(3) and A(4).
   1.5, 2, 1.5, 0
2. Where is A largest? What is f there?
   At x = 2, where f(2) = 0.
3. $f(t) = 2 \sin t$: compare the slope of A with the height of f at x = 1.6 and at x = 4.
   Both ≈ 2.00 at x = 1.6, and both ≈ –1.51 at x = 4.
4. Keep x the same and change a from 0 to 1. What changes and what stays the same?
   A(x) shifts down by $\int_0^1 f(t)\,dt$ (a constant). Its slope, f(x), stays the same.

3 Record use the tool and fill in the table

Accumulation values

Round to 2 decimal places.

f(t)	a	x	A(x)	f(x) = slope of A	A increasing or decreasing?
2 – t	0	3	1.5	–1	decreasing
2 sin t	0	2	2.83	1.82	increasing
piecewise	0	1	–1	–1	decreasing
piecewise	0	3	0	2	increasing
2 – t	1	3	0	–1	decreasing

4 Explain what did you discover?

1. A′(x) = f(x) .
2. A increases where f is positive and decreases where f is negative .
3. A has a maximum where f changes from positive to negative .
4. Changing a shifts (moves up or down) the graph of A but does not change its slope .

5. Use the Fundamental Theorem of Calculus to explain why $\frac{d}{dx}\int_1^x t^2\,dt = x^2$.

   The slope of an accumulation function at x is the height of the integrand at x, so the derivative is x^2. (Check: the integral is $\frac{x^3}{3} - \frac{1}{3}$, whose derivative is x^2.)

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5 Apply a new situation, no tool

1. $F(x) = \int_0^x (t^2 - 4)\,dt$. Find $F'(3)$.

   $3^2 - 4 = 5$

2. $G(x) = \int_2^x \cos t\,dt$. Where on $[0, 2\pi]$ does G have a local maximum?

   $x = \frac{\pi}{2}$, where cos t changes from positive to negative.

3. If $\int_0^5 f(t)\,dt = 12$ and $\int_0^2 f(t)\,dt = 4$, find $\int_2^5 f(t)\,dt$.

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4. Find $\frac{d}{dx}\int_0^{x^2} \sin t\,dt$.

   $2x \sin(x^2)$, by the chain rule.

Teacher notes

Intended discovery: The accumulation function $A(x) = \int_a^x f(t)\,dt$ increases where $f > 0$, decreases where $f < 0$, and its slope is $f(x)$ ($A' = f$); changing a shifts A by a constant.

Likely misconceptions:

- A is largest where f is largest (it is largest where f changes from + to −).
- Confusing A(x) (an area) with f(x) (a height).
- Thinking the lower limit changes the derivative.

Answers that can vary: Readings for 2 sin t are rounded; any x within 0.1 of the exact extremum is acceptable in the Try this list.